Target validation paths — demo-nsclc-egfr-l858r-post-osi-d3m0

The diagnostic and biomarker workup that hardens the targetable-feature call

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Libby is an experimental decision-support tool. The recommendations in this report have not been reviewed by a clinician treating this patient. Do not act on this report without consulting a qualified oncologist.

Target validation paths

EGFR L858R, MET amplification (GCN 8.2), and T790M-absent are all confirmed at decision-relevant resolution, so no diagnostic test gates the rank-1 trial. The validation work here is about resistance characterization at progression, not target confirmation — the goal is to make sure the case is interpreted as MET-amplified bypass resistance rather than something more complicated.

EGFR L858R

The most consequential miss at this point would be SCLC histologic transformation, which is the highest-PPV resistance pattern that no liquid biopsy can confirm. Tumor NGS that includes TP53 and RB1 status is the screen — co-loss is the strongest predictor of transformation, with reported rates of 3–14% across post-osimertinib cohorts. If imaging morphology shifts or co-loss is detected, fresh re-biopsy with a neuroendocrine IHC panel (chromogranin, synaptophysin, INSM1, Ki-67) is the only way to confirm. ctDNA covering EGFR C797S and exon-20 inserts is the parallel on-target resistance check; a positive C797S result reframes the next-line conversation toward fourth-generation EGFR-TKIs or combination strategies.

MET amplification

The FISH GCN 8.2 result clears SAFFRON's enrollment threshold (≥ 6) cleanly, but MET amplification can be focal and primary-vs-metastatic discordance is a documented confounder in this setting. A second-site FISH (or matched archival block) refines confidence that the dominant disease — not just the biopsied site — is amplified. MET IHC (clone SP44) is an orthogonal-modality complement when comparing to the SAFFRON eligibility profile, and a comprehensive NGS panel that picks up HER2 / BRAF / FGFR co-alterations frames durability expectations and the next-line plan.

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.